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4R/2025 PHY4204C

2025

PHYSICS

(Major-II)

Paper : PHY4204C

(Quantum Mechanics and Applications)

Full Marks : 45

Time : Two hours

The figures in the margin indicate full marks for the questions.

1. Answer the following questions : $1 \times 4 = 4$

(a) Which one of the following does not support quantum nature of radiation ?

(i) Photoelectric effect

(ii) Compton effect

(iii) Diffraction

(iv) Black body radiation

(b) What is the probability of finding a particle represented by wave function $\Psi(r, t)$ in unit volume ?

(c) Write the operator form of momentum in quantum mechanics.

(d) If $\hat{A} = \frac{d^2}{dx^2}$ and $\Psi(x) = ae^{2x}$, then find the eigenvalue of $\Psi(x)$ for operator $\hat{A}$.

2. Answer the following questions : $2 \times 3 = 6$

(a) Illustrate briefly on wave particle duality.

(b) Find the uncertainty of momentum of an electron confined in a $10 \text{ \AA}$ width box.

(c) Prove that $[\hat{x}, \hat{P}_x] = i\hbar$.

3. Answer the following questions : $5 \times 3 = 15$

(a) Write the physical significance of wave function.

Mention the properties of well-behaved wave function. $2 + 3 = 5$

- (b) An electron is confined within a region of width 1.0×10^{-10} m. Estimate the uncertainty in the momentum of the electron. Given $\hbar = 1.05 \times 10^{-34}$ J-Sec.

OR

- (c) The work function of potassium is 2eV . When ultraviolet light of $3500 \text{ \AA}$ wavelength is incident on the potassium surface, what will be the maximum energy of the emitted photoelectron?

- (d) Write the Einstein's photoelectric equation. Explain laws of photoelectric effect from this equation. $2+3=5$

OR

- (e) Derive time independent Schrödinger's equation. 5

4. Answer **any one** from the following questions: $10 \times 1 = 10$

- (a) An incident X-ray photon is scattered by free electron at rest through an angle θ , using relativistic expression of electron energy, show that the change in the wave length of the photon is given by

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

Where m_0 rest mass of electron, h Planck's constant, c Velocity of light.

10

OR

(b) What is de Broglie's hypothesis?

Describe briefly Davisson-Germer experiment to verify de Broglie's hypothesis.

2+8=10

5. Answer **any one** from the following questions :

10×1=10

(a) A particle of mass m is confined in a one dimensional infinite potential well of potential

$$V(x) = 0 \quad \text{for } 0 < x < L$$

$$= \infty \quad \text{for } 0 \geq x \geq L.$$

Find the energy of the particle in the n^{th} quantum state. Also find the normalized wave function of the particle for n^{th} energy state.

5+5=10

OR

- (b) Explain Heisenberg Uncertainty principle. Verify Uncertainty principle by γ -ray microscope experiment.

$$2+8=10$$